Comparative Analysis of ML Algorithms for Detecting Intrusion Using the NSL-KDD Dataset

Ravindra Singh Koranga
School of Computer Science and Engineering
Galgotias University
Greater Noida, India
333koranga@gmail.com

Arun Kumar Rai
School of Computer Science and Engineering
Galgotias University
Greater Noida, India
arunrai.iitism@gmail.com

Shobhit Kumar
School of Computer Science and Engineering
Galgotias University
Greater Noida, India
kumar.shobhit05@gmail.com

Hradesh Kumar
School of Computer Science and Engineering
Galgotias University
Greater Noida, India
hradeah.pachauri@gmail.com

2025 3rd International Conference on Disruptive Technologies (ICDT) | 979-8-3315-1958-2/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICDT63985.2025.10986545

***Abstract*— With the advent of computer networks, it has become very convenient to communicate, share and collaborate with others in a short time. All the online services depend on computer networks. With the rise of computer networks there is a increase in threats of cyberattacks. Intrusion by an attacker is very common which can put the whole network at risk. Thus Intrusion detection has become very important today. There are several traditioanl methods which can indetify the intrusion. This study focuses on the use of Machine Learning to detect intrusion. Several machine learning models are trained on the NSL-KDD dataset from Kaggle. This dataset contains several factors which can result in intrusion. These factors are input to Machine Learning model to make prediction about any attack in the network. A correlation analysis of these factors is also done to identify the relationships among these factors. The performance of several Machine Learning algorithms are analysed based on several parameters and the best model is determined.**

Keywords—Intrusion Detection, Machine Learning, NSL KDD

I. Introduction

A computer Network represents a collection of computers which are connected to each other. These share information with each other which can lead to possibility of intrusion in the network. To detect intrusion Machine Learning techniques are used. These techniques are based on model trained on the NSL-KDD dataset to detect intrusion.

ML models offers several advantage over any traditional method. The ML algorithms can effectively analyse the input and detect any types of intrusion using the data. This is done in a very efficient manner as compared to traditional method. The changing patterns of attacks can be identified using these methods. Thus ML is very dynamic and can adapt to any variability in the attacks. The attackers may be very dynamic and can change the signature methods to avoid any types of detection, but with the help of ML the detection becomes more resilient to new and unknown attacks. These allows to effectively and efficiently identify any types of intrusion.

There are some challenges as well when using ML for intrusion detection. These challenges must be overcome to effectively identify the intrusion attempts by the attacker. These challenges can be overcome by using large amount of suitable dataset and properly pre-processing the data before training.

II. Literature Review

Today, the role of network is very crucial as it allows all the IT infrastructure to be run properly. With the increasing size of network, there is a rise in cyber-attacks as well.

There is several existing research in the field of cyber security. This section analyses the existing research work been explored in the field using various Machine Learning techniques. [1] has proposed an innovative way of detecting the intrusion patterns. These techniques provide an insight in to the impact of intrusion.

Another study by Alshebeili et al. [2] has developed a very effective deep learning model which allows the the detection of intrusion using extensive pattern analysis. The study has shown that certain deep learning models are more effective. This is due to the reason that these models rely on heavy use of data processing which allows them to effectively find patterns in the data. [3] has worked on the several factors which can result in the intrusion being carried out by the attacker.

[4] has worked on several deep learning methods to determine intrusion which allows them to achieve accuracy of 97.4% which is quite impressive considering the variability in the data. Similarly, [5] has given a method which relies on hybrid approach to achieve an accuracy of 99.2% in the given data. This makes the model very effective. [6] has used probabilistic networks to find the intrusion. [7] has worked on methods like information Gain and relief selection. In [8] has used other deep learning methods which significantly enhance the detection of intrusion in an efficient manner.

[9] has used a multi-level SVM for the task. [10] compares how Gaussian NB differs from other models for intrusion detection.

III. Dataset Description

The study used The NSL-KDD data for our work. The dataset was created by researchers at the National Aerospace Laboratories (NSL).

The NSL-KDD dataset contains a total of about 148517 connection records and 42 features, including information about the source and destination of the connection, protocol type, service, flags, and various other characteristics. The Data contains several Protocols and different types of services. Fig.

Authorized licensed use limited to: SARDAR VALLABHBHAI NATIONAL INSTITUTE OF TECH. Downloaded on March 01,2026 at 11:21:57 UTC from IEEE Xplore. Restrictions apply.

1 shows protocol type pie chart distribution and Fig. 2 shows Services pie chart distribution in the data.

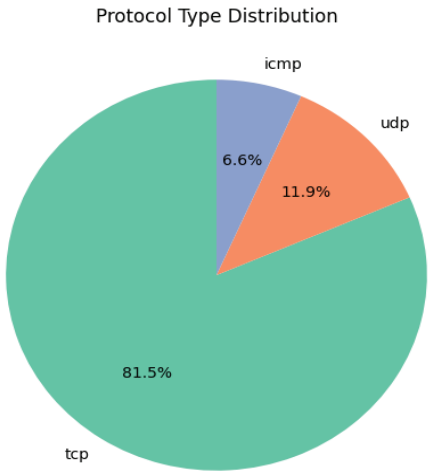

Fig. 1. Protocol Type Distribution in Pie Chart

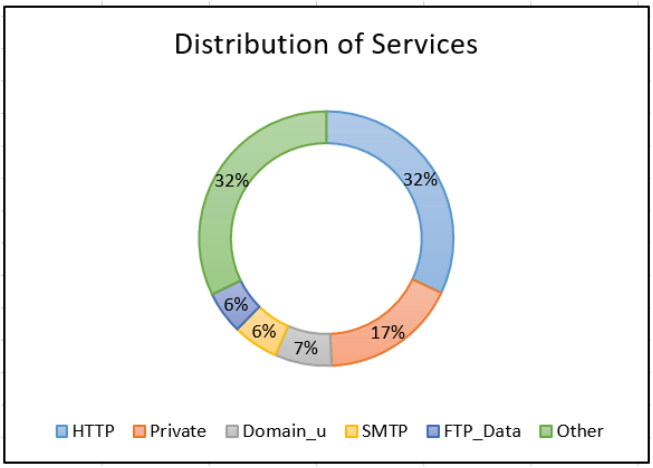

Fig. 2. Distribution of Services

There is several existing research in the field and in the use of machine learning application in intrusion. Some techniques have become quite popular for intrusion detection as the dataset contains several numerous features which can help in determining intrusion patterns.

IV. Methodology

A. Feature Selection

One of the initial steps is to analyse the provided dataset. A common technique used for this purpose is correlation analysis, which helps in identifying the relationship between multiple variables. The heat map depicted in Fig. 3 presents the correlation matrix among the input variables and the attack variable. Once the correlation analysis is complete, the input data is analyzed. To analyze the data feature selection is done. There are several different flags in the dataset. Fig. 4 shows the count of each flag type. Next step involves carefully extracting features from the given dataset which is analyzed and input to ML models. Using the analysis effort is made to enhance the model's performance by mitigating overfitting, increasing generalization, and improving interpretability.

Feature selection can be done through several methods. It helps to remove unnecessary information from the data. Feature selection can offer an advantage as it helps to filter out the redundant input. Filter methods is a technique which finds the correlation between target variable and feature. Only those features that pass the statistical test are selected for use in the model. On the other hand, in wrapper methods the subset of features is considered for performance using a particular machine learning algorithm. The appropriate feature which pass the performance is selected. The Embedded method ensures that feature selection is performed during the training.

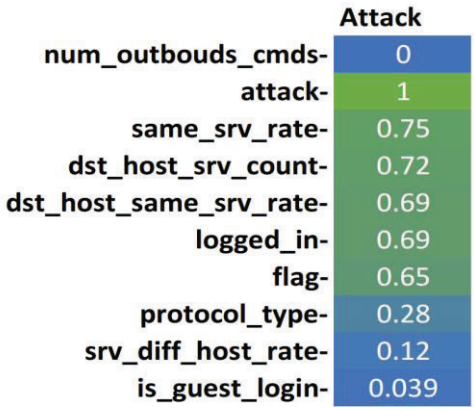

Fig. 3. Heat map of Attack label and other values

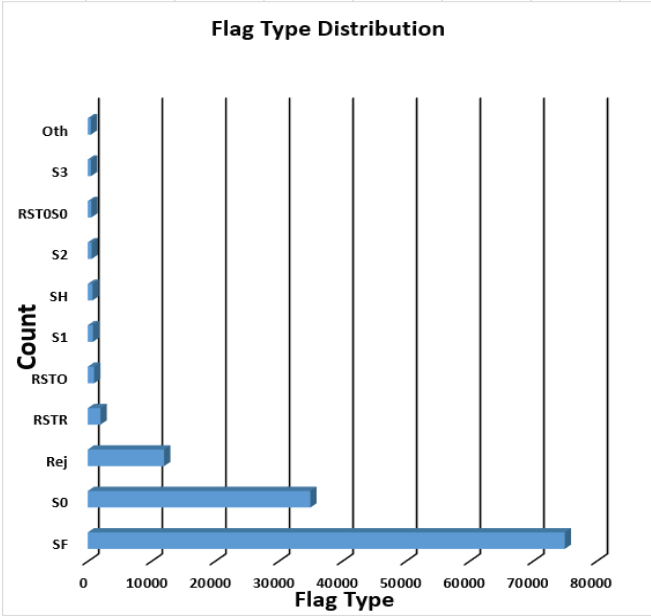

Fig. 4. Count of different Flag types in the data

To determine the list of important features, we utilized Extra Tree Classifier from the scikit-learn library. This is an ensemble method for feature selection and classification, also known as Extremely Randomized Trees. The algorithm is an extension of Random Forest, another popular ensemble method for classification and regression tasks, that is implemented in the scikit-learn library. These algorithms are the main focus of this study.

To prevent overfitting and improve overall performance, the Extra Trees Classifier algorithm constructs numerous decision trees on a dataset and then averages the results. At each split in the tree, the algorithm randomly selects a subset of features, which is why it is known as "extremely randomized". One of the main advantages of using the Extra Trees Classifier for selection of feature as it has the ability to

Authorized licensed use limited to: SARDAR VALLABHBHAI NATIONAL INSTITUTE OF TECH. Downloaded on March 01,2026 at 11:21:57 UTC from IEEE Xplore. Restrictions apply.

capture the relationships among the variables and it can identify the importance of the input which highly affect prediction. The feature importance are obtained by averaging the feature importance of all decision trees in the ensemble. Based on the comparison of importance of input features by this method and on doing the post analysis we have dropped these features from the data.

- num_outbound_cmds
- land
- is_host_login
- urgent
- num_shells
- su_attempted
- num_access_files

B. *Training ML models*

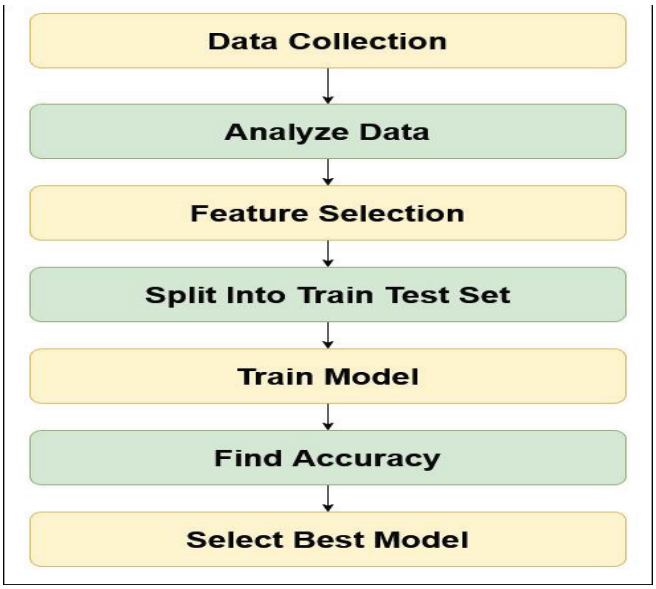

Fig. 5. Flowchart of Intrusion Detection Model Training and Selection

Machine learning (ML) has become a popular mode of intrusion detection. The dataset is analysed by the model to identify any patterns of intrusion. There are several models that can be applied on the dataset to detect intrusion. Fig. 5 shows the steps taken while performing the task of training and testing the models. We have explored following models in our work

V. Result and Conclusion

The accuracy score of each model in the test data is given as follows :

DecisionTreeClassifier	99.74%
LogisticRegression	83.13%
SVC	98.64%
AdaBoostClassifier	99.81%
GradientBoostingClassifier	98.36%
RandomForestClassifier	99.83%
KneighboursClassifier	98.46%
GaussianNaiveBayes	42.80%
MultilayerPerceptron	96.49%

Fig. 6 shows the plot of accuracy of various models. We find that Decision Tree, Ada Boost and Random Forest Classifiers give best result for our data. Whereas, Logistic Regression and Gaussian Naïve Bayes do not perform so well.

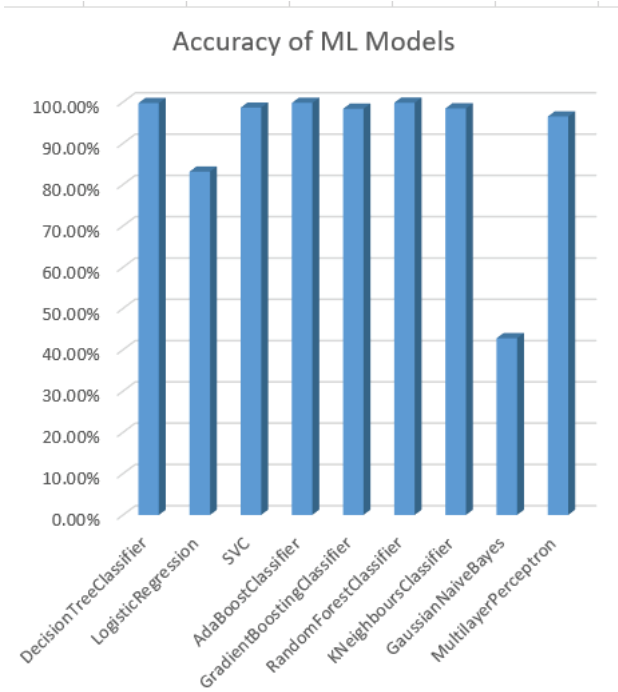

Fig. 6. Plot showing the accuracy of different ML Models

k-Nearest is a popular machine learning model which is known for its simplicity and robustness. The model analyses the k nearest samples to determine the output. Naive Bayes makes a strong independence assumption between the input features. It is fast and can handle large datasets, but its performance can be affected by the independence assumption.

Some other powerful algorithms for intrusion detection are Support Vector Classifier which uses a support vector plane, Random Forest Classifier which uses probabilistic modelling, Gradient Boosting Classifier which is also probabilistic based, AdaBoost Classifier, and Multi-Layer Perceptron. These algorithms are capable of handling both linear and non-linear decision boundaries. However, such algorithms are high computational requirement and may need parameter tuning.

These algorithms are compared based on several different performance parameters. The algorithm with best performance will be selected for use. The best performing algorithm will be the one with highest efficiency and accuracy. These parameters define the best suited for the task.

References

[1] Sabillon, "A Comprehensive Cybersecurity Audit Model to Improve Cybersecurity Assurance: The CyberSecurity Audit Model (CSAM)," *2017 International Conference on Information Systems and Computer Science (INCISCOS)*, Quito, Ecuador, 2017, pp. 253-259, doi: 10.1109/INCISCOS.2017.20..

[2] Alshebeili et al. Intrusion detection system based on deep learning. In 2018 8th International Conference on Computer and Information Technology (ICCIT).

[3] AlMahmeed et al., "Zero-day Attack Solutions Using Threat Hunting Intelligence: Extensive Survey," *2022 International Conference on*

Authorized licensed use limited to: SARDAR VALLABHBHAI NATIONAL INSTITUTE OF TECH. Downloaded on March 01,2026 at 11:21:57 UTC from IEEE Xplore. Restrictions apply.

Data Analytics for Business and Industry (ICDABI), Sakhir, Bahrain, 2022, pp. 309-314, doi: 10.1109/ICDABI56818.2022.10041568..

[4] Alrifai et al., "Deep Learning-based Intrusion Detection System," Journal of Network and Computer Applications, vol. 140, pp. 10-18, 2019.

[5] Chen et al., "A Hybrid Deep Learning-based Intrusion Detection System," Journal of Ambient Intelligence and Humanized Computing, vol. 11, pp. 9977-9988, 2020.

[6] Dong et al., "Real-Time Network Intrusion Detection System Based on Deep Learning," 2019 *IEEE 10th International Conference on Software Engineering and Service Science (ICSESS)*, Beijing, China, 2019, pp. 1-4, doi: 10.1109/ICSESS47205.2019.9040718..

[7] Guo et al., "Research and Design of Electric Power Engineering Project Management System Bsed on Blockchain Technology," *2022 International Conference on Blockchain Technology and Information Security (ICBCTIS)*, Huaihua City, China, 2022, pp. 80-84, doi: 10.1109/ICBCTIS55569.2022.00029

[8] Li et al., "Research and Implementation on the Operation and Transaction System Based on Blockchain Technology for Virtual Power Plant," *2022 International Conference on Blockchain Technology and Information Security (ICBCTIS)*, Huaihua City, China, 2022, pp. 165-170, doi: 10.1109/ICBCTIS55569.2022.00046.

[9] Lacoste et al., "Flying over Mobile Clouds with Security Planes: Select Your Class of SLA for End-to-End Security," *2013 IEEE/ACM 6th International Conference on Utility and Cloud Computing*, Dresden, Germany, 2013, pp. 50-59, doi: 10.1109/UCC.2013.25..

[10] Kumar et al., "Software Fault Prediction using Wrapper based Feature Selection Approach employing Genetic Algorithm," *2022 OPJU International Technology Conference on Emerging Technologies for Sustainable Development (OTCON)*, Raigarh, Chhattisgarh, India, 2023, pp. 1-7, doi: 10.1109/OTCON56053.2023.10113911.

Authorized licensed use limited to: SARDAR VALLABHBHAI NATIONAL INSTITUTE OF TECH. Downloaded on March 01,2026 at 11:21:57 UTC from IEEE Xplore. Restrictions apply.